Hazem Mohsen Abdelaziz Mohamed

Junior Robotics Research Engineer

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Professional Summary

Junior Robotics Research Engineer with hands-on experience in autonomous systems, cooperative autonomous vehicles, SLAM, robotics perception, and sensor fusion. Experienced in developing research-oriented robotics software pipelines using ROS 2, MATLAB, Python, and C++ for localization, control, navigation, and risk-aware autonomous driving applications. Strong background in simulation-based validation, real-time system integration, stereo vision, and intelligent transportation research with practical exposure to scalable autonomous vehicle architectures and robotics experimentation.

Research Interests

Cooperative Autonomous Vehicles • Smart Transportation Systems • Intelligent Robotics • Sensor Fusion • SLAM and Localization • Autonomous Navigation • Robotics Perception

Education

Bachelor of Science in Mechatronics Engineering Sept 2019 – July 2024

Ain Shams University, Cairo, Egypt

GPA: 3.09 / 4.00

Graduation Project: Autonomous Upper Humanoid Robot Grade: A+

Research & Experience

Part-Time Autonomous Vehicles Testing Engineer – Micro Engineering Tech Inc.

June 2026 – Present

- Design, execute, and validate scenario-based test cases for autonomous vehicle software in simulation environments, covering both nominal and edge-case driving conditions.
- Perform functional, integration, and system-level testing to evaluate perception, localization, planning, and vehicle control modules under diverse traffic scenarios.
- Analyze simulation outputs, identify software defects and unexpected vehicle behaviors, document test results, and collaborate with development teams to improve system performance and reliability.
- Develop and maintain structured test cases, test procedures, and validation reports to support autonomous vehicle verification and validation (V&V) activities.
- Contribute to continuous improvement of testing workflows, ensuring comprehensive coverage of safety-critical driving scenarios and autonomous driving requirements.

Part-Time Research Assistant – Multi-Robot Systems (MRS) Lab, German University in Cairo

Oct 2025 – Jan 2026

- Developed the core software infrastructure for the *Vector_V1* autonomous vehicle research platform with focus on modularity, scalability, and maintainability.
- Designed and optimized ROS 2 communication architectures enabling efficient multi-module data exchange across perception, control, and teleoperation pipelines.
- Implemented modular teleoperation and control systems with integration support for Logitech G920 steering wheel interfaces.

- Produced technical documentation for deployment workflows, software architecture, validation procedures, and system integration.

Robust VSLAM Using MATLAB Mobile Streaming – MathWorks Challenge #213

Sept 2024 – Nov 2024

- Developed a MATLAB App Designer-based framework supporting multi-mode Visual SLAM experimentation and robotics perception workflows.
- Implemented monocular Visual SLAM pipelines for indoor and outdoor environments using feature-based localization and mapping techniques.
- Integrated IMU-camera sensor fusion strategies to improve trajectory stability and localization robustness.
- Designed GPS-assisted outdoor pose estimation modules for mobile robotics experimentation.

Fuzzy-Logic Based Fusion for Autonomous Vehicle Localization – MathWorks Challenge #20

Mar 2024 – Aug 2024

- Designed adaptive localization pipelines integrating LiDAR, IMU, GPS, and camera data for autonomous vehicle positioning.
- Modeled and validated multi-sensor fusion strategies using MATLAB and Simulink for autonomous driving applications.
- Evaluated localization reliability under diverse driving conditions using synthetic simulation environments and KITTI datasets.

Graduation Project Intern – Quanttronix

Oct 2023 – Jan 2024

- Contributed to the development of an autonomous upper humanoid robot platform integrating navigation, manipulation, and low-level control systems.
- Designed wiring architectures, actuator control interfaces, and power distribution systems for reliable robotics hardware integration.
- Implemented low-level control algorithms supporting arm manipulation, grasping, and autonomous movement capabilities.

Technical Projects

Stereo Vision TTC Estimation for Autonomous Scaled Vehicles

- Designed stereo vision pipelines for depth estimation and Time-To-Collision (TTC) calculation in scaled autonomous vehicle environments.
- Implemented depth extraction, object distance estimation, and perception validation using Webots simulation.

ROS 2 Autonomous Vehicle Software Architecture

- Built modular ROS 2 software architectures supporting autonomous navigation, inter-module communication, and scalable robotics deployment.
- Developed real-time robotics pipelines integrating control, localization, and perception modules.

Lane Detection and Computer Vision-Based Driving Assistance

- Implemented lane detection and feature extraction pipelines using OpenCV, edge detection, Hough transforms, and image enhancement techniques.
- Evaluated real-time visual perception performance using autonomous driving video datasets.

MATLAB Mobile Sensor Fusion and Localization Applications

- Developed MATLAB-based mobile robotics applications integrating GPS, IMU, camera, and mobile sensor streaming.
- Implemented real-time localization visualization, pose estimation, and SLAM experimentation interfaces.

Publications & Ongoing Research

Safety-Aware Cooperative Control for Autonomous Scaled Vehicles Using Risk Assessment-Based Decision Making

Ongoing research work focused on cooperative autonomous vehicle safety, risk assessment-based control architectures, Webots simulation environments, and experimental hardware validation.

Certifications

Robust VSLAM Using MATLAB Mobile Streaming
State Estimation and Localization for AVs
C Programming from Basics to Mastery
OpenCV Bootcamp
Code Foundation for ROS
Stateflow Onramp

Technical Skills

Robotics & Autonomous Systems: ROS 2, SLAM, Sensor Fusion, Localization, Autonomous Navigation, Forward and Inverse Kinematics, Manipulation Systems, Robotics Perception

Programming: MATLAB, Python, C++, C

Computer Vision: Features Detection & Tracking, Lane Detection, Visual Odometry

Simulation & Development Tools: MATLAB, Simulink, Carla, Webots, Gazebo, RViz, CoppeliaSim, Linux, Git, Docker

References

Omar M. Shehata

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Research Supervisor – Multi-Robot Systems Lab

Roberto G. Valenti

Principal Robotics Researcher, MathWorks

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Research Supervisor – MathWorks Challenges

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Graduation Project Supervisor